

Foundation of Blockchain Technology (IS501)

Pre-Reading Material – Lecture 3

Block Structure & Merkle Tree

Estimated Reading Time: 20–25 minutes

Estimated Reading Time: 15–20 minutes

Course Context

This lecture introduces the internal structure of a Bitcoin block and explains how blocks are organized within the blockchain. You will learn the components of a **block header**, the role of **hash pointers**, **Merkle trees**, and the **nonce** in Proof of Work. The lecture also discusses **block versions**, **difficulty representation using nBits**, and the organization of serialized blocks. Together, these concepts explain how Bitcoin ensures the integrity, security, and immutability of its blockchain.

Why This Topic Matters

Every blockchain is built from blocks, and every block contains information that links it securely to the previous block. Understanding block headers, Merkle trees, and cryptographic hashing is essential for learning how blockchains detect tampering, verify transactions efficiently, and achieve consensus. These concepts form the foundation for Bitcoin mining, transaction validation, and blockchain security.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Identify the fields present in a Bitcoin **block header**.
- Explain the purpose of the **previous block hash**, **Merkle root**, **timestamp**, **nBits**, and **nonce**.
- Describe how a **Merkle tree** is constructed from transaction hashes.



- Explain the role of **Proof of Work** in block validation.
- Interpret the purpose of the **nBits** field and its relationship to mining difficulty.
- Differentiate between a **block header** and a **serialized block**.

Activate Your Prior Knowledge

Before reading further, think about the following questions.

1. How can a blockchain detect if someone modifies a transaction that was recorded years ago?
2. Why doesn't a block simply store all transactions inside its header instead of using a Merkle root?
3. If thousands of miners are trying to create the next block simultaneously, how does the network decide which block becomes valid?

Prerequisite Concepts

Before studying this lecture, you should be familiar with:

- Cryptographic hash functions (SHA-256)
- Basic blockchain terminology
- Digital transactions
- Binary and hexadecimal representation
- Previous concept of immutable ledgers

Core Concepts

1. Bitcoin Block Header

Every Bitcoin block begins with an **80-byte block header**, which is used during the Proof-of-Work process.

The block header contains six fields:

- **Version** – Specifies the block validation rules.
- **Previous Block Hash** – Stores the SHA-256 hash of the previous block header, linking blocks together.
- **Merkle Root** – Represents a single hash computed from all transactions in the block.



- **Timestamp** – Records when the miner created the block header.
- **nBits** – Stores the compact representation of the mining target (difficulty).
- **Nonce** – A value that miners repeatedly change until the generated block hash satisfies the required target.

2. Previous Block Hash

Each block stores the hash of the previous block header.

This creates a chain of blocks where modifying any previous block changes its hash, thereby invalidating all subsequent blocks.

This mechanism provides the blockchain with **tamper resistance**.

3. Merkle Tree

A Merkle tree is constructed using the **transaction IDs (TXIDs)** of all transactions contained in a block.

The construction process is as follows:

- Arrange all transaction hashes in order.
- Pair adjacent hashes.
- Hash each pair together using SHA-256 twice.
- Repeat the process until only one hash remains.

The final hash is called the **Merkle Root**, which is stored in the block header.

If any transaction changes, the Merkle Root also changes, immediately indicating that the block has been modified.

4. Proof of Work and Nonce

Miners repeatedly modify the **nonce** while hashing the block header.

The objective is to generate a block hash that is **less than or equal to the target threshold**.

Finding such a nonce demonstrates that computational work has been performed, which is known as **Proof of Work (PoW)**.

5. Target Threshold and nBits

The mining difficulty is represented in a compact format called **nBits**.

Instead of storing the complete 256-bit target value, Bitcoin stores a compressed representation.

The lower the target value, the more difficult it is to find a valid block hash.

6. Serialized Block

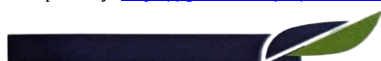
A serialized Bitcoin block consists of:

- Block Header
- Transaction Count
- List of Transactions

The first transaction must always be the **coinbase transaction**, which collects the mining reward and transaction fees.

Guided Reading Questions

1. Why does every block contain the **hash of the previous block**?
2. What is the purpose of the **Merkle Root**?
3. Why do miners continuously change the **nonce** during mining?
4. What information is stored in the **block header**?
5. Why is the **coinbase transaction** always the first transaction in a block?



Reflection Activity

Consider a blockchain containing one million blocks.

1. What would happen if a hacker modified a transaction in Block 500?
2. Which field in the block header would immediately become invalid?
3. Why would the attacker need to recompute the Proof of Work for subsequent blocks?

Write a **150–200 word reflection** explaining how the previous block hash and the Merkle root together help maintain blockchain integrity.

Self-Assessment

Fill in the Blanks

1. The Bitcoin block header has a fixed size of _____ bytes.
2. The _____ stores the hash representing all transactions in a block.
3. Miners repeatedly change the _____ to obtain a valid block hash.
4. The _____ field stores the compact representation of the mining target.
5. The first transaction in every Bitcoin block is called the _____ transaction.

True or False

1. Every transaction is stored directly inside the block header.
2. If one transaction changes, the Merkle Root also changes.
3. The previous block hash links one block to another.
4. The nonce is modified during the mining process.
5. The block header contains the complete list of transactions.

